Descriptive + Time-Series Trend Analysis (Excel / Laravel + Chart.js)

1. Data Requirements (Keep it Simple)

Students only need a **sales transaction table**:

Minimum fields:

- `date`
- `product_name`
- `category` (coffee, pastry, etc.)
- `quantity`
- `price`
- `total_sales`

Optional (if available):

- `time` (for hourly trends)
- `payment_method`

2. Overall Process (Step-by-Step)

Step 1: Data Collection

- Export POS data OR manually encode sample data
- Store in:
  - Excel / Google Sheets (simplest), OR
  - MySQL (if using Laravel project)

Step 2: Data Cleaning

- Remove duplicates
- Fix missing values
- Standardize product names (important!)

☞ This step is often ignored—but critical.

Step 3: Aggregation (Core of Trend Analysis)

Convert raw data into summaries:

Examples:

- Daily sales
- Weekly sales
- Monthly sales
- Sales per product

☞ SQL Example:

```
SELECT DATE(date) as day, SUM(total_sales) as sales
FROM transactions
GROUP BY day;
```

Step 4: Visualization (Where Trends Appear)

Use simple charts:

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- Line chart → sales over time
- Bar chart → top-selling products
- Pie chart → category distribution

Tools:

- Excel (easiest)
- Chart.js (for web systems)
- Power BI (optional but powerful)

Step 5: Identify Trends (Interpretation)

Students should answer:

- Which days have highest sales?
- What products are consistently popular?
- Are sales increasing or decreasing?
- Peak hours? (if time data exists)

Step 6: Simple Forecasting (Optional but Valuable)

Keep it basic:

Option A: Moving Average

- Smooths fluctuations

- Easy to compute in Excel or code

Option B: Linear Trend Line

- Shows growth/decline

☞ Excel does this automatically → **Add Trendline**

Example Insights

- “Milk tea sales increased by 15% over 3 months”
- “Weekends generate 40% higher revenue”
- “Product X is declining—possible replacement needed”